

Lab Guide — Lakeflow Connect & Orchestration

Work top to bottom. Everything lives in **your** schema inside the `dev-sh-training` catalog. Set the `schema` widget to your own name on every notebook before you run it.

- Catalog (fixed): `dev-sh-training`
- Your schema: whatever you set in `00_setup` (e.g. `jane_doe`)
- Landing volume: `/Volumes/dev-sh-training/<you>/landing/`

⚠ How to use every notebook in this lab — read first. The widgets (`schema`, `scenario`, ...) are defined in the **first cell**. You must **run that first cell** before anything else so the widget boxes appear at the top of the notebook. Then fill in your `schema` (and any other options) and run the remaining cells (or **Run all**). If you skip the first cell, the widgets never appear, and the notebook falls back to the placeholder default and stops with an error.

The value you type is substituted everywhere automatically — via `${schema}` in SQL notebooks and via the widget in Python notebooks.

Setup (once)

1. Add this repo as a **Git folder** (Workspace → Create → Git folder, URL `https://github.com/realboom/lakeflowconnect`).
2. Open `notebooks/00_setup`, **run the first cell** so the `schema` widget appears, set it to your name, then **Run all**. You now have a `schema` and a `landing` volume with `filedrop/` and `sftp/` subfolders.

Part A — COPY INTO (batch file ingestion)

Concept: `COPY INTO` loads *new* files from a folder into a table and is **idempotent** — rerun it and already-loaded files are skipped. Simple, batch, great for “a file shows up, pull it in.”

1. Open `generate_data`, set `schema`, `scenario` **Part A - file drop (COPY INTO)**, `num_rows` = 25, **Run all**. → writes a comma CSV to `landing/filedrop/incoming/`.
2. Open `part_a_copy_into`, set `schema`, **Run all**. → creates `bronze_file_eligibility` and loads the file.
3. **Run the COPY INTO cell again.** It loads **0 new rows** — that’s idempotency (it remembers what it already ingested).

4. Generate another Part A file (step 1 again) and re-run COPY INTO → only the **new** file is loaded.

✅ Takeaway: COPY INTO = the simplest incremental batch ingest.

Part B — Auto Loader (“SFTP” source, mimicked on a Volume)

Concept: A vendor normally drops files on **SFTP**. We mimic that by landing pipe-delimited files in `landing/sftp/incoming/` and let **Auto Loader** (`cloudFiles`) ingest them incrementally. Auto Loader keeps a **checkpoint** of what it has processed and is built for continuously-arriving files.

1. Open `generate_data`, scenario **Part B – SFTP source (Auto Loader)**, `num_rows = 25`, **Run all.** → writes a **pipe-delimited** CSV to `landing/sftp/incoming/`.
2. Open `part_b_autoloader`, set `schema`, **Run all.** → creates `bronze_sftp_eligibility` and ingests the file. The last cells show the data.
3. Generate another Part B file and re-run Part B → only the **new** file is picked up (checkpoint).

✅ Takeaway: Auto Loader = incremental, checkpointed ingestion — the standard for flat-file / “SFTP” feeds. Contrast with Part A: COPY INTO is batch SQL; Auto Loader is a streaming source with schema evolution + rescue (next).

Part C — Auto Loader resilience

Same `bronze_sftp_eligibility` pipeline from Part B. We show the two ways Auto Loader absorbs drift. **Do rescued data first** (it runs on the original schema), then schema evolution (it permanently adds a column).

C1 — Rescued data (bad value, original schema)

1. `generate_data` → scenario **Part C – rescued data (SFTP)** → **Run all.** Half the rows have a bad `eff_date` of `N0T-A-DATE`. (`eff_date` is typed `DATE` via `schemaHints`.)
2. `part_b_autoloader` → **Run all.** The bad values can’t cast to `DATE`, so Auto Loader **rescues** them: the row still ingests with `eff_date = NULL`, and the original value is parked in `_rescued_data`.
3. The last cell shows the rescued rows.

💡 A malformed value didn’t crash the pipeline or get silently dropped — it’s captured for audit/backfill.

C2 — Schema evolution (a new column)

1. `generate_data` → scenario **Part C - schema evolution (SFTP)** → **Run all**. The file has a new `risk_tier` column.
2. `part_b_autoloader` → **Run all**. Auto Loader detects `risk_tier`. In a notebook it **stops once** to record the new column — if you see an “unknown field”/schema-change message, **just run the cell again** and it succeeds, with `risk_tier` added to the table (existing rows `NULL`).
3. `SELECT risk_tier, count(*) ... GROUP BY risk_tier` to confirm.

🔔 A new field appeared and the table evolved on its own. (In Part E a task **retry** makes this one-run automatic.)

Part E — Orchestration (Lakeflow Pipelines + Jobs)

Now we orchestrate the eligibility data you just created. You’ll build **one pipeline** and a few **jobs** by hand.

Where you build these — read this once. In the Databricks **left sidebar**, find **Jobs & Pipelines**. Tip: **right-click it and open in a new browser tab**, so you can keep this guide open side-by-side. On the **Jobs & Pipelines** page, click the blue **Create** button (it has a dropdown). The dropdown lets you pick what to build — **ETL Pipeline** or **Job**. Each step below tells you which one to choose: - **E1** builds an **ETL Pipeline** (that’s what a Spark Declarative Pipeline is). - **E2, E3, E5** build **Jobs**.

E1 — The silver pipeline (an ETL Pipeline / Spark Declarative Pipeline)

This builds a streaming `silver_eligibility` table from your Auto Loader bronze (`bronze_sftp_eligibility`).

1. **Jobs & Pipelines** (left sidebar) → blue **Create** button → **ETL Pipeline**. You land in the pipeline editor on a new pipeline named like *New Pipeline 2026-...*, with a starter file `transformations/my_transformation.py`.
2. **Rename the pipeline:** click its name at the top-left and change it to `silver_eligibility_<you>`.
3. ⚠️ **Set the default catalog + schema — this is how the pipeline finds *your* tables.** Open the pipeline **Settings** (gear icon, or the `:` menu → *Settings*) and set:
 - **Default catalog = `dev-sh-training`**
 - **Default schema = *your* schema** (the one from `00_setup`)
 - **Serverless** compute

(The editor shows a default like `... / default` near the top-right — you must change it to yours.) Because the code uses the bare name `bronze_sftp_eligibility` (no catalog/schema), it resolves against this default catalog + schema — so your pipeline reads *your* bronze and writes *your* silver. Same code for everyone; only the default schema differs.

4. **Add the transformation SQL.** The starter file is Python, but our logic is SQL:

- Right-click `my_transformation.py` → **Rename** to `silver_eligibility.sql` (or click + to add a new `silver_eligibility.sql` and delete the empty `.py`).
- Open `notebooks/silver_eligibility` in your Git folder, copy the `CREATE OR REFRESH STREAMING TABLE silver_eligibility ...` statement, and paste it into the file.
- (Optional — AI authoring: instead of pasting, click **Create with Genie Code / Generate** and paste this prompt — then compare what it writes to the SQL in the repo before running: > “Create a streaming table in SQL named `silver_eligibility` that reads incrementally from the > streaming table `bronze_sftp_eligibility`. `initcap` `first_name` and `last_name`; cast `eff_date` and > `term_date` to DATE; keep `member_id`, `plan_code`, `line_of_business`; add `silver_loaded_at` as > the current timestamp. Use `FROM STREAM bronze_sftp_eligibility` so it's incremental, not a reload.”)

5. Click **Run** (▶ Run pipeline) → it builds `silver_eligibility` in your schema from `bronze_sftp_eligibility`. (Part B must have run first so your bronze table exists.)

E2 — Table-update trigger (event-driven)

Make the silver pipeline run **automatically when new bronze data lands**. 1. **Jobs & Pipelines** → blue **Create** button → **Job**. Add one task: - Task name `refresh_silver`, Type **Pipeline**, select your silver pipeline. 2. **Add a trigger** on the job → type **Table update** → table `dev-sh-training.<you>.bronze_sftp_eligibility`, condition **ANY_UPDATED**. (Set min time between triggers = 60s.) 3. Test it: run **generate_data** (Part B) then **part_b_autoloader** so new rows land in bronze. Within ~a minute the job fires on its own (run origin: “Triggered by table update”) and refreshes silver.

💡 The trigger baselines when you create it and fires on **new** commits — so make a fresh bronze load after you set it up.

E3 — The DAG: gold → report → notify (with a conditional branch)

Create a multi-task job. 1. **Jobs & Pipelines** → blue **Create** button → **Job**, name it `eligibility_gold`. 2. Task **gold** — Type Notebook → `notebooks/gold_eligibility_summary`, param `schema` = your schema. Under **Advanced** → **Retries**, set **Max retries = 2**. 3. Task **report** — Type **Notebook** → `notebooks/eligibility_report`, param `schema` = your schema. **Depends on gold**. 4. Task **notify_pass** — Notebook `notebooks/notify`, param `result` = `pass`. Depends on `report`. **Run if = All succeeded**. 5. Task **notify_fail** — Notebook `notebooks/notify`, param `result` = `fail`. Depends on `report`. **Run if = At least one failed**. 6. **Run** the job → `gold` → `report` → `notify_pass` (and `notify_fail` is skipped). Green path.

E4 — Repair / partial re-run (recover from a schema change)

A realistic failure: someone renames a column upstream and the downstream report breaks. 1. Open `gold_eligibility_summary` and rename the output column `member_count` → `enrolled_members`, save. 2. **Run** `eligibility_gold`. → `gold` **succeeds** (table rebuilt with the new name), but `report`

fails: UNRESOLVED_COLUMN ... member_count ... Did you mean 'enrolled_members'?, and notify_fail runs. 3. **Fix** eligibility_report: change member_count → enrolled_members, save. 4. On the failed run click **Repair run**. → **gold is skipped** (already built), only report + notify_pass re-run → success.

🔔 You don't rerun the expensive upstream that already succeeded — fix and resume from the failure. (To reset for a repeat: revert both files back to member_count.)

E5 — For each (fan-out + concurrency)

Run a task once per line of business, in parallel. 1. **Jobs & Pipelines** → blue **Create** button → **Job** eligibility_by_lob. Add a **For each** task. - **Inputs:** ["Commercial", "Medicaid", "Medicare Advantage", "Individual"] - **Concurrency:** 4 2. **Add a task to loop over** → Notebook notebooks/for_each_lob, params schema = your schema and lob = {{input}}. 3. **Run** → four iterations run in parallel, each building gold_elig_<lob>.

Concurrency FAQ (you'll be asked): concurrency is **1–100** (default 1); effective parallelism = $\min(\text{concurrency}, \# \text{ inputs})$; the workspace cap is **2,000 concurrent task runs**. Size it to what your downstream can handle, not the max. See the instructor runbook for the full FAQ.

E6 — Retries

Notice the **Max retries = 2** you set on the gold task. Retries heal transient failures — and they're exactly what makes the Part C **schema-evolution** handshake complete in a single automated run (Auto Loader stops once to record the new column; the retry picks it up). Put a retry on any task that ingests evolving files.

Clean up / start over

Run **notebooks/reset** (set your schema) to drop the lab tables and clear the landing folders. Your schema and volume stay. Re-run **generate_data** to begin again.